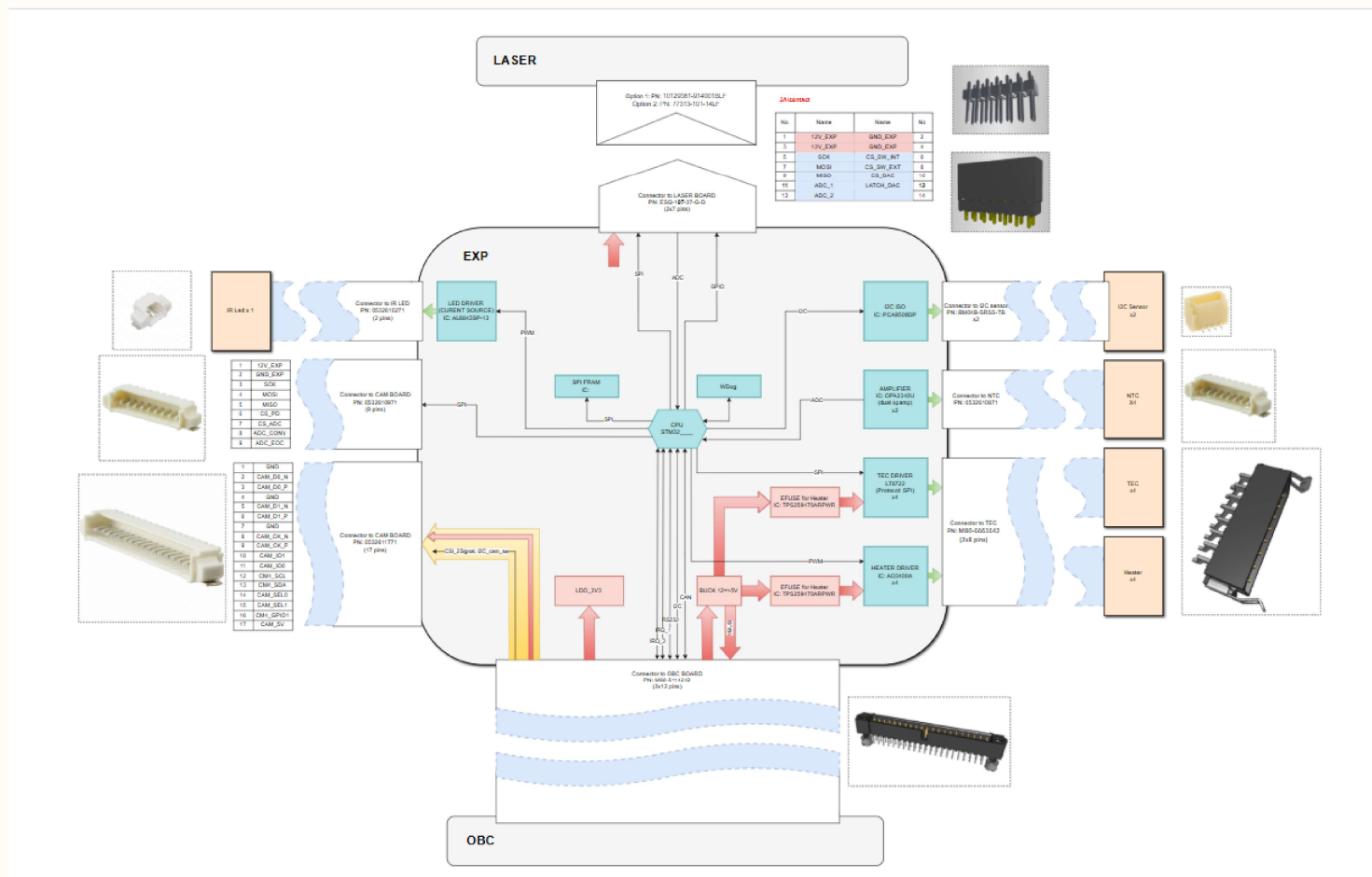
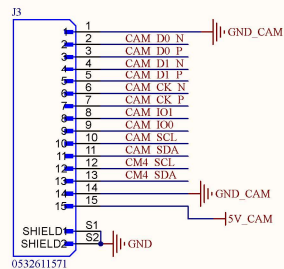
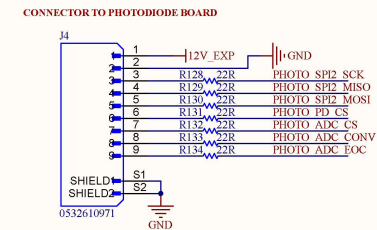
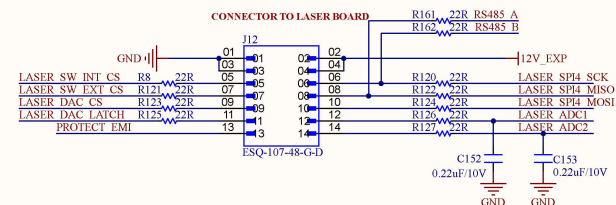
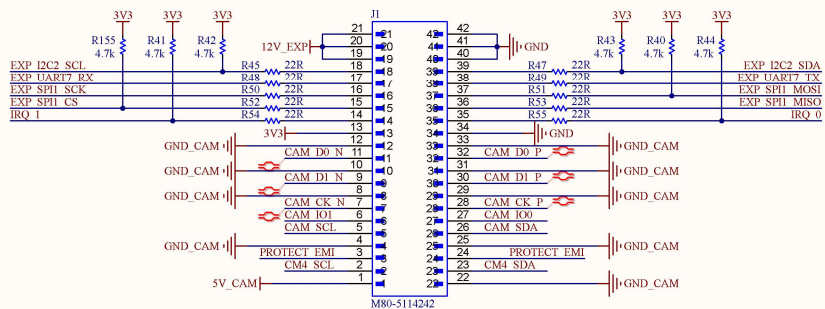


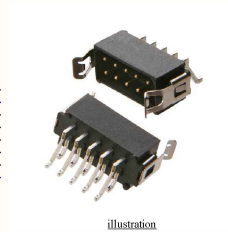
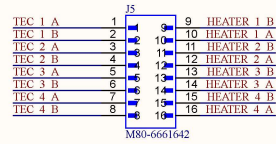


Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: Block_Diagram.SchDoc	Sheet 1 of 12
Checked by: BAO BUI Q.	Document No: 1	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: HEADER_1.SchDoc	Sheet 2 of 12
Checked by: BAO BUI Q.	Document No: 2	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025

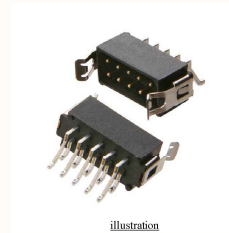
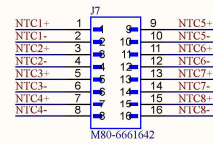
CONNECTOR TO TEC & HEATER



Harwin Inc. M80-6661642

- **ELECTRICAL:**
CURRENT RATING: 2.2A/CONTACT
VOLTAGE RATING: 800V/CONTACT
- **ENVIRONMENTAL:**
TEMPERATURE RANGE: -55°C TO +125°C
- **PHYSICAL:**
PITCH-MATING: 0.079" (2.00mm)
- **WIRE:**
WIRE SIZE: 24 AWG
MAX CURRENT: 2A per Wire
ISULATION MATERIAL: PTFE
- **MATING:**
Harwin Inc. M80-8881606

CONNECTOR TO NTC



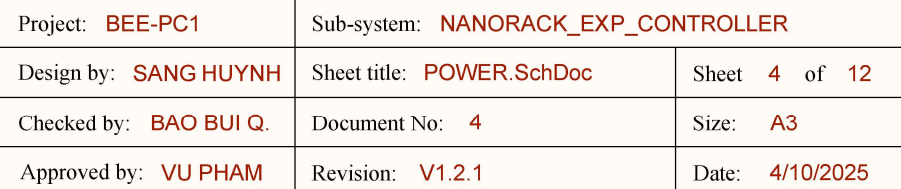
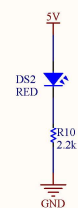
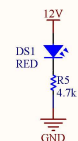
Harwin Inc. M80-6661642

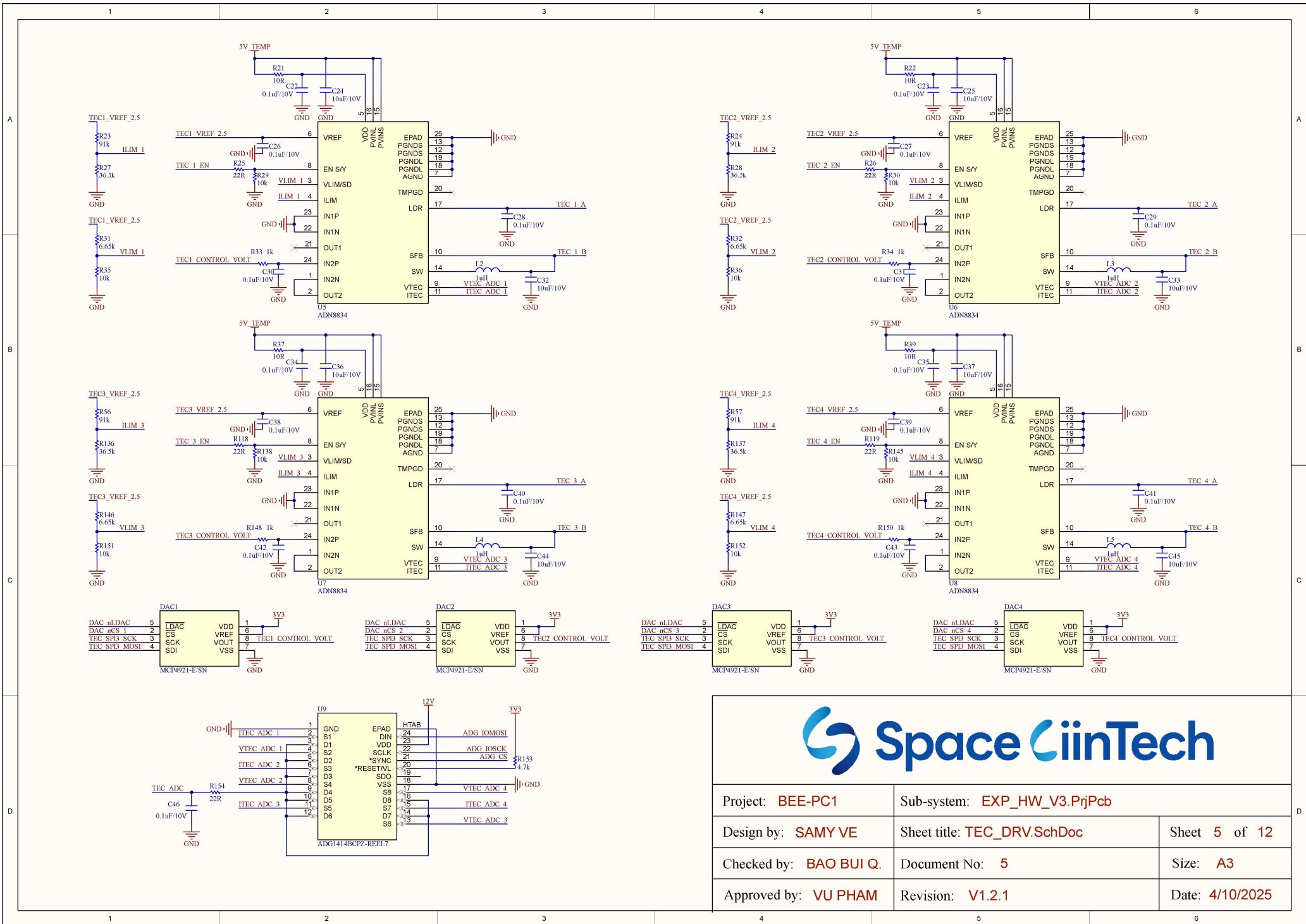
- **ELECTRICAL:**
CURRENT RATING: 2.2A/CONTACT
VOLTAGE RATING: 800V/CONTACT
- **ENVIRONMENTAL:**
TEMPERATURE RANGE: -55°C TO +125°C
- **PHYSICAL:**
PITCH-MATING: 0.079" (2.00mm)
- **WIRE:**
WIRE SIZE: 24 AWG
MAX CURRENT: 2A per Wire
ISULATION MATERIAL: PTFE
- **MATING:**
Harwin Inc. M80-8881606

MOUNTING HOLE

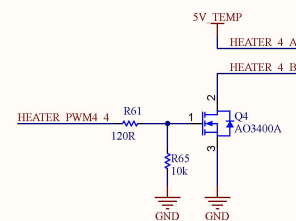
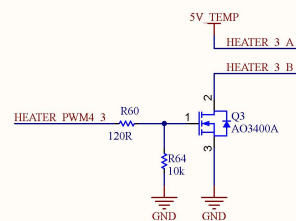
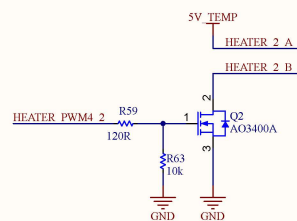
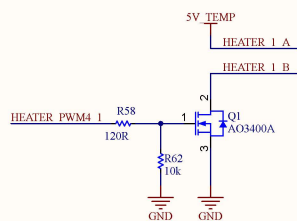


Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: HEADER_2.SchDoc	Sheet 3 of 12
Checked by: BAO BUI Q.	Document No: 3	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025

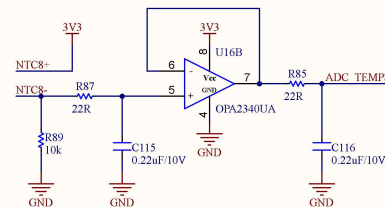
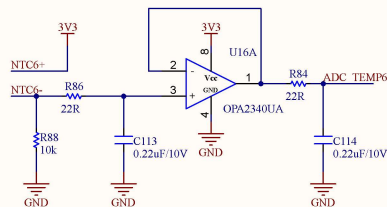
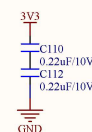
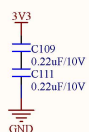
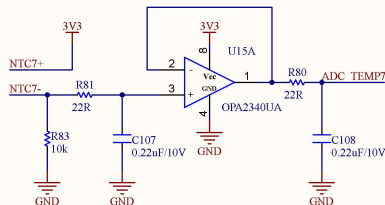
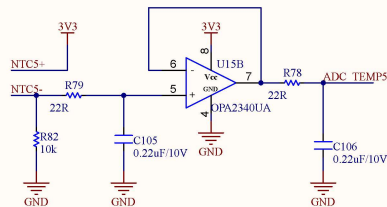
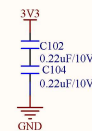
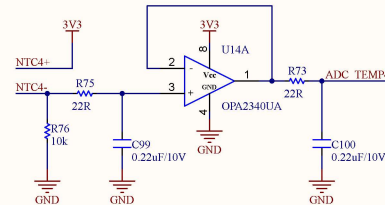
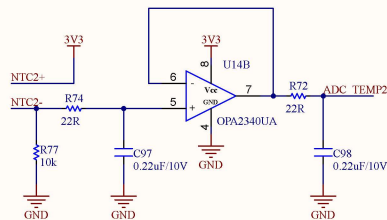
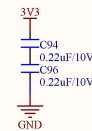
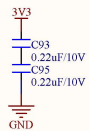
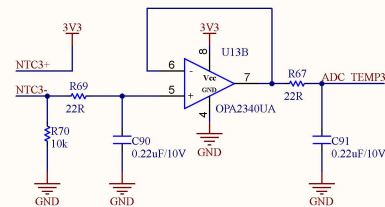
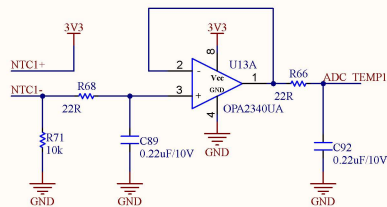
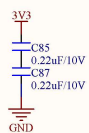




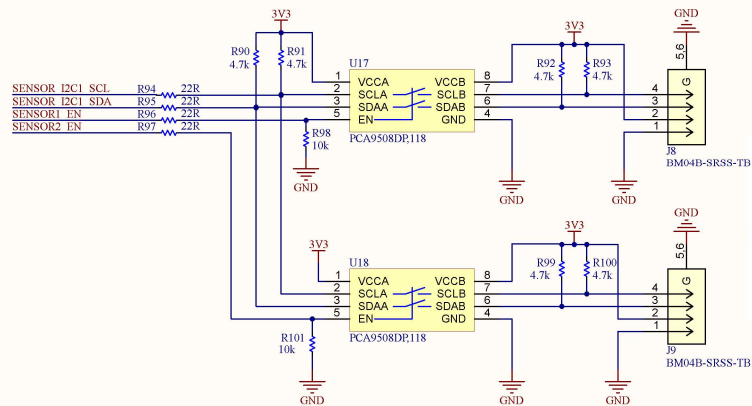
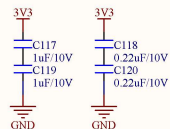
Project: BEE-PC1	Sub-system: EXP_HW_V3.PrjPcb	
Design by: SAMY VE	Sheet title: TEC_DRV.SchDoc	Sheet 5 of 12
Checked by: BAO BUI Q.	Document No: 5	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



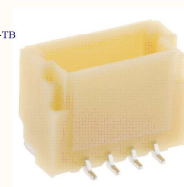
Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: HEATER.SchDoc	Sheet 6 of 12
Checked by: BAO BUI Q.	Document No: 6	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



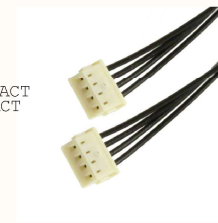
Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: NTC.SchDoc	Sheet 7 of 12
Checked by: BAO BUI Q.	Document No: 7	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



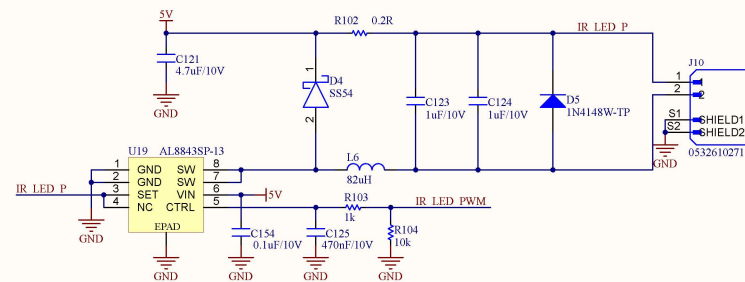
JST Sales America Inc. BM04B-SRSS-TB



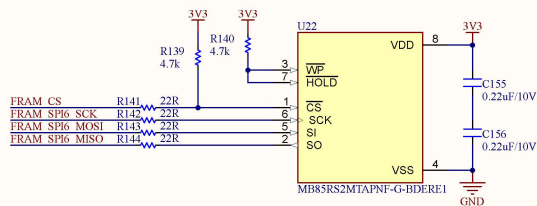
- **ELECTRICAL:**
CURRENT RATING: 1.0A AC/DC/CONTACT
VOLTAGE RATING: 50V AC/DC/CONTACT
- **ENVIRONMENTAL:**
TEMPERATURE RANGE: -25°C ~ 85°C
- **PHYSICAL:**
PITCH-MATING: 0.039" (1.00mm)
- **WIRE:**
WIRE SIZE: 26 AWG
MAX CURRENT: 0.1A per Wire
ISULATION MATERIAL: PTFE
- **MATING:**
JST Sales America Inc.
A04SR04SR30K152B



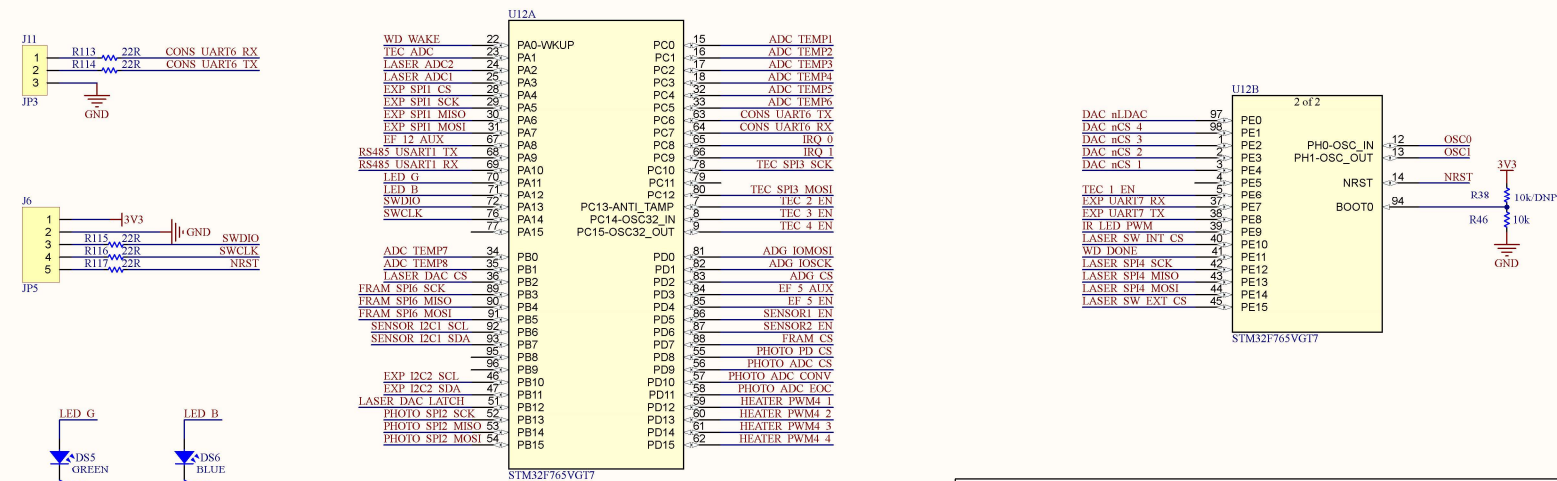
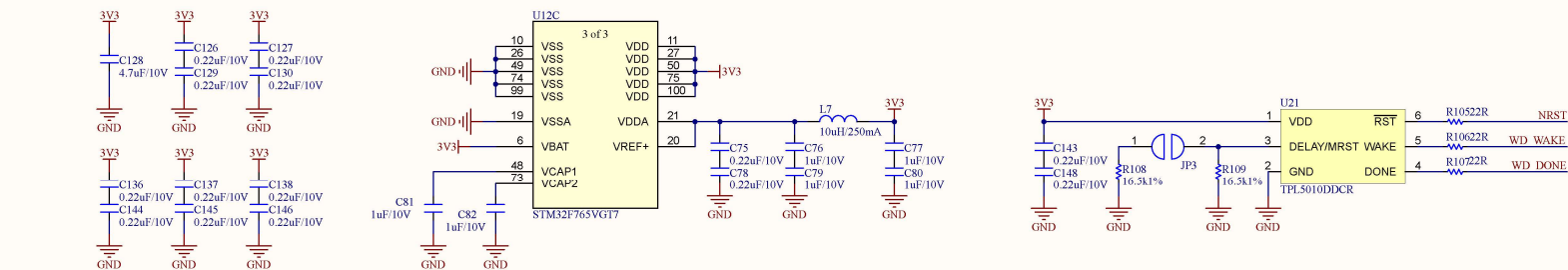
Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: I2C_Sensor.SchDoc	Sheet 8 of 12
Checked by: BAO BUI Q.	Document No: 8	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



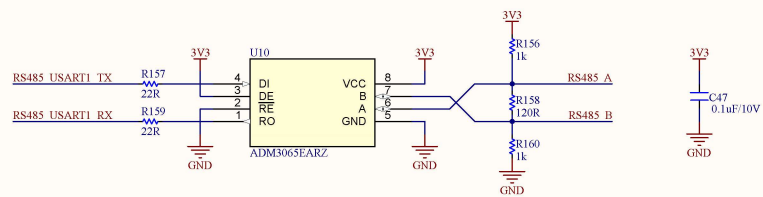
Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: IR_LED.SchDoc	Sheet 9 of 12
Checked by: BAO BUI Q.	Document No: 9	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: FRAM.SchDoc	Sheet 10 of 12
Checked by: BAO BUI Q.	Document No: 10	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



Project: BEE-PC1	Sub-system: NANORACK_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: MCU.SchDoc	Sheet 11 of 12
Checked by: BAO BUI Q.	Document No: 11	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025



Project: BEE-PC1	Sub-system: EXP_HW_V3.PrjPcb	
Design by: SAMY VE	Sheet title: RS485.SchDoc	Sheet 12 of 12
Checked by: BAO BUI Q.	Document No: 12	Size: A3
Approved by: VU PHAM	Revision: V1.2.1	Date: 4/10/2025